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biotope inc

Tokyo based, biotope, is a strategic design firm that assists innovators through all stages of developing a vison from idea to reality.

Through fusing design, marketing, and organization development methodologies for cooperative creation, we design ecosystems where the ideas of innovators are realized, create communities for co-creation, and support in all stages of bringing ideas to reality.

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Translator

Aleksander Songe-Møller Soleim

The non-designer's guide to design thinking

What a marketer learned in design school

SASO KUNITAKE

Dear creators of the future,

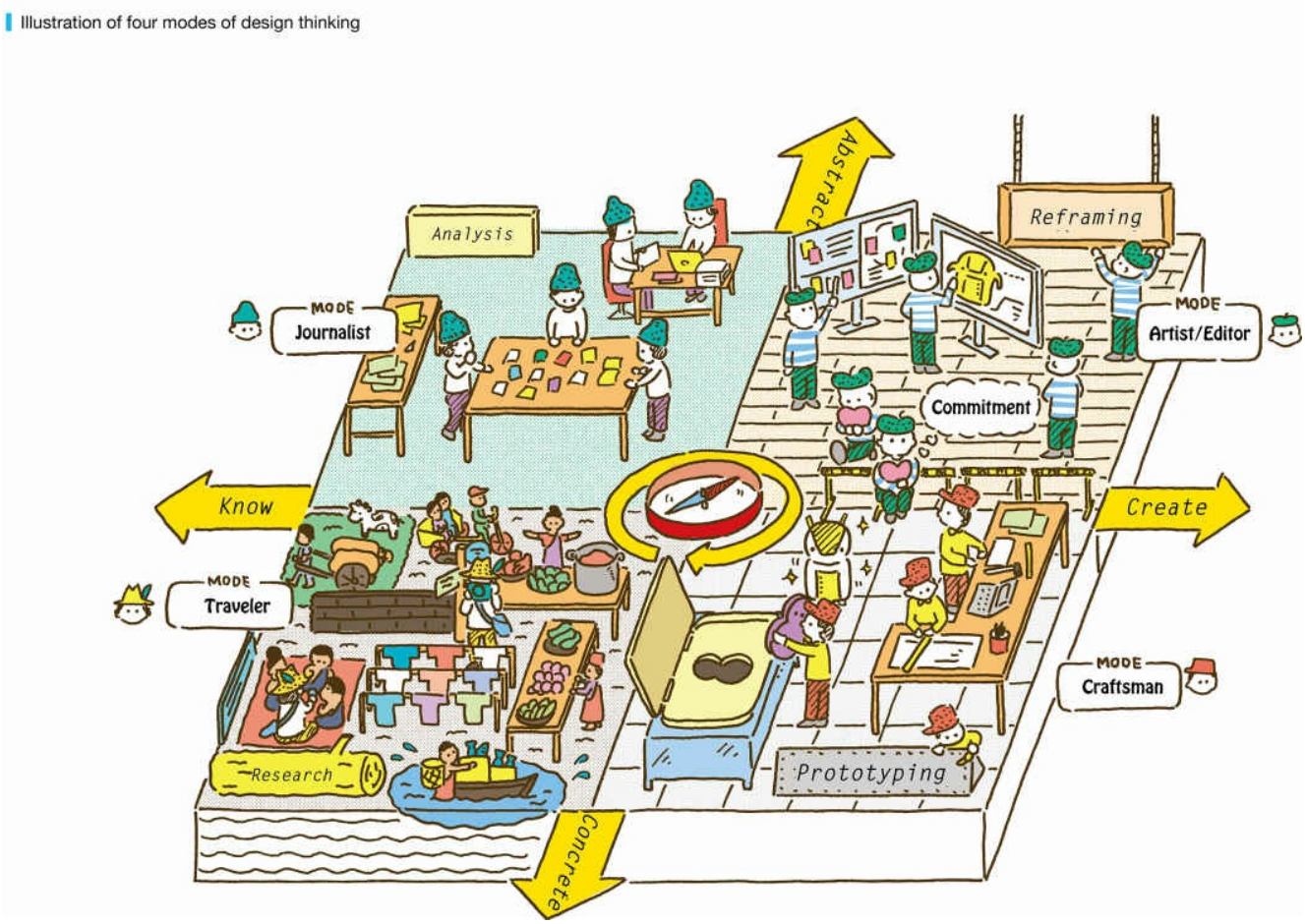
If you aim at being the first to reach the top of an unexplored mountain,
Then say goodbye right now to your sheltered, comfortable environment,
throw yourself into the harshest creative competition in the world.

Polish the three intellects: philosophical, technological, and societal
by constructive thinking through thorough critique and interdisciplinary competition.
Train in the three powers of survival: the power to stick out, the power to lay roads, and the power to build mountains on your own.

Saso's book will surely be a reassuring compass for the first chapter of your voyage.
The second chapter and beyond, depends on your power to build mountains on your own.
I wish you the best of luck on your voyage.

Hiroshi Ishii (Professor at MIT Media Lab)

Illustration of four modes of design thinking



Index

Contents

Preface

Chapter 0 America: An Advanced Education Style for the 21st Century

Chapter 1 Learning Hybrid Intellectual Production from Designers

Chapter 2 Creator-Spirit School

Chapter 3 A Compass for Creative Problem Solving

Chapter 4 The Switch to Creative Production Mode

Chapter 5 Design as a Career in the Business World

Chapter 6 Design Thinking is a Life Skill that Lets You Live a Fulfilling Life

Afterword

Commentary

Preface

Lately, one of the most popular subjects at top MBA programs in the US, including Stanford and Harvard, is design. I can already hear some people saying "Huh? Business professionals study design?" However, learning design to advance your business career has become common at top American and European schools.

While MBAs teach how to manage business efficiently using logical thinking as a base, design teaches how to create a completely new business, product, service, or process that goes beyond linear growth of existing business. I earned a graduate degree of design method from Illinois Institute of Technology's design school (hereafter ID: https://www.id.iit.edu/), a top-tier school for design thinking education; design which also business professionals can study. ID is a legendary school that originated from Bauhaus in the area of human centered design. During my school days, friends from around the world who were enrolled in MBA programs listened with great interest about my design studies at ID. There is a reason for this interest. Industries around the world are now seeking to restructure and reform, and a movement to not just rebuild existing business, but create new kinds of business, is blossoming. There is currently a trend at the world's top MBA schools to adopt design concepts into their curriculum.

After studying design at ID, I returned to Japan to work at Sony. There, I took part in such activities as building a new company-wide incubation eco-system known today as Sony Seed Acceleration Program for creating new business using design thinking and lean startup. Since then, I have firmly been a part of the world of design innovation, but before ID I had never studied design. At university I majored in law, and started my career as a marketer at P&G. There, I learned MBA-like marketing and management skills such as data analysis, statistical forecast, and concept development. My trigger into the world of design was my

P&G marketing mentor, who successfully launched Febreze in the Japanese market. He has a rare capability of creating something from nothing by changing the rules of game. He was gifted at redefining (in other word rebuilding) brands that were declining, coming up with completely new concepts, and launching them as new brands. But I could not understand how he did this. I was good at logic and analysis, but even if I could steadily run existing brands and even double their sales, no matter how hard I tried, I was not able to come up with the kind of ideas that could create something from nothing. Using analytical and logical thinking to create new concepts was not sufficient, so I started my journey into the creative world in search of methods to generate new concepts.

Design is often associated with creators and artists, and it is not a very familiar concept for the great majority of business people. The word "design" combines the meanings of "creation" and "realization". As a methodology for creative problem solving, which discovers intrinsic problems not obvious at first glance and solves them creatively. It is now considered comparable to the strategy consulting methodology used in strategic consulting firms like McKinsey. Although many business professionals have a hard time grasping design, what I learned in D school were design thinking methods for creating new value that are actually systematized well enough that anyone can learn them. IDEO/Stanford Design School has issued many resources for businesses to utilize design thinking concepts. However, I found design thinking to be much broader and deeper. These are important areas for business people to understand.

In the design world there is a large volume of knowledge that business professionals can utilize to create new value in their current jobs. Design does not necessarily contradict how existing business is being run; the two can be utilized side by side. In this book I will introduce the overall picture of design thinking mastery. This is what I learned during my adventure at design school from the view

point of non-designer. I use my experience to present a road map for how to become a business leader with a design-mind, who can actively solve tangible and intangible problems of society in creative ways; and also provide a compass to navigate this road map, in the form of a concrete methodology. By presenting the contents of my learnings from design school, I also hope to give an overview of how to put design thinking into practice.

Nothing would honor me more than to have business professionals and entrepreneurs who read this book, gain insight on how to be design-minded business leaders in the 21st century.

chapter 0

America: An advanced education style for the 21st century

"65 percent of children entering grade school this year (2011) will end up working in careers that haven't even been invented yet".

Cathy N. Davidson, professor at Duke University

Global Trends Surrounding White Collar Workers

IBM is now the world's largest design firm. They employ more than one thousand designers from all over the world. This change is symbolic of the recent interest in design which is now spreading broadly, and has become relevant for everyone. To explain why there is so much buzz about design thinking lately, I would like to introduce the global trends behind its expansion. For white collar professionals who make intellectual output their livelihood, creative problem solving has now become an essential skill. I have had many opportunities to visit American and European design schools, MBA schools, and design firms to have discussions with global innovators and leaders. Through these experiences I noticed four important trends:

Trend 1: Businesses in Developed Countries Need Disruptive Innovation

The maturing of societies in developed countries have led corporations to make less revolutionary innovation in order to grow their business. As a result, they face a situation where they must struggle for their survival. Competition with the conglomerates of emerging economies with start-ups is getting tougher, and so existing businesses need to boldly transform their profit models. "Our core business is disappearing in front of us." "If we don't start doing something new, we won't survive." In many companies, such remarks are becoming common. At the same time, knowledgeable workers are facing competition with the cheap but decent-quality workforce of emerging economies. When forced to compete in the same market, salaries can only go one way - down. The only way for them to survive is to create value that only they can create, and thus put themselves in a unique position.

In Asia, the "technological innovation myth" used to prevail in Japan: Innovation

and value was supposed to be created under the technological leadership of research and development departments. But in the coming age, the only way for developed countries to differentiate themselves from emerging giants like China and India is to make use of their latent technology and culture to create original value.

Trend 2: Preparing For a Creative Society Where Everyone is a Creator

The arrival of the Internet, smartphones, and social network services (SNS) completely changed the structure of society and business. Creative expression used to be limited to a small portion of creative workers, but today, anyone can publish creative content with just a smartphone and a computer. Facebook has a global user base of 1.3 billion people, and the photo-sharing service Instagram has reached 300 million.(As of Aug, 2013) An example of everyday creativity is photo/movie sharing on Instagram or Facebook. People actively compose and edit their photos in order to gain more "likes". In Japan, an amateur parody video of the Japanese idol group AKB48's dance video, "Koi Suru Fortune Cookie", was a big hit on YouTube. Young people all over Japan imitated their dance and uploaded their versions to Youtube. In this day and age, constructing your own website, utilizing your SNS community base, and marketing skills to make a living off of it has become commonplace.

As for important changes in the near future, desktop fabrication techniques, such as 3D printing and laser CNC, are quickly becoming widely available technology. Soon anyone will be able to design items to meet everyday needs. It will become possible to use your own creativity to design your own life or, if you feel up to it, start a business. It will not only be possible for an individual to become a small-scale creator and entrepreneur, but also have the capability to turn creations into a side-business, or even the primary source of livelihood.

Trend 3: The Competition for Jobs Between Machines and Humans

It has been predicted that artificial intelligence technologies, such as deep learning, will develop remarkably quickly in the next 20 years. As a result of these innovations, many jobs currently performed by humans will be taken over by machines. According to a study designed to identify what occupations will disappear as a result of artificial intelligence development, published by professor Osborne of Oxford University, the following two types of work are difficult to substitute with artificial intelligence:

1. Work that requires communication skills: such as healthcare, education, and psychological counseling.
2. Creative fields such as design, engineering, and management.

As artificial intelligence evolves, computers will be able to analyze many types of data, understand the patterns, and give efficient answers based on data analysis. As a result, jobs focusing on analysis are considered to be easily substituted by artificial intelligence. On the other hand, creating new solutions and making decisions based on the machine's analysis is likely to remain human work for now.

Trend 4: Finding Happiness With What You Have

In the matured societies of developed nations, more and more people are seeking spiritual satisfaction over materialistic satisfaction for a fulfilling life.

The world population is estimated to reach 9 billion by 2050. With limited resources to be divided between more people, the 21st century may be all about learning to be content with what you have.

During my design school days, I discovered that using creativity every day can lead to flow and happiness. When each individual designs his or her lifestyle

according to his or her preferences or inclinations, this leads to an increase in happiness. In order to do this, learning how to unleash your own creativity will become important.

Creativity in US Academia and Business

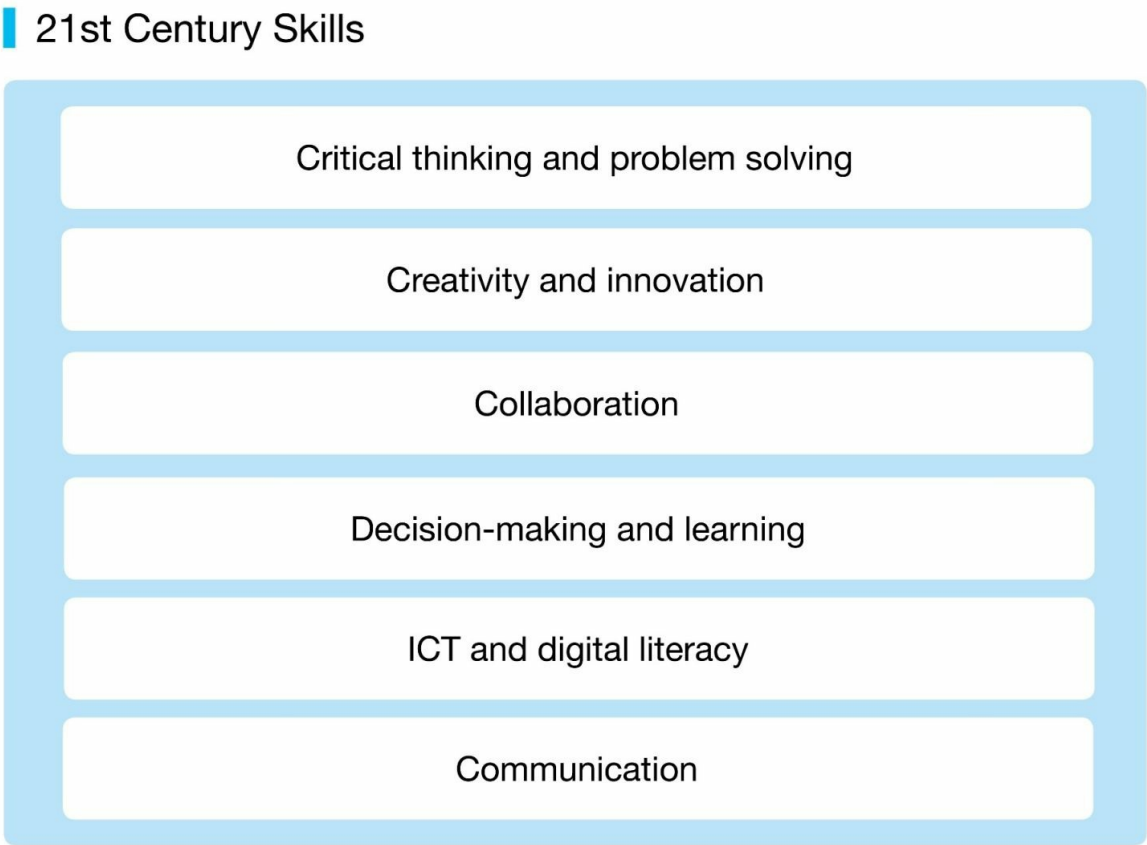
In the midst of these trends, the country that first successfully incorporated creativity into industry was USA. Since 1970, Stanford University has researched how engineers can adopt the designer's way of thinking via the concept of "visual thinking". In Chicago during 1980's, user interface development merged with visual thinking resulting in the emergence of human centered design. Later in the 1990s, design strategy consulting industry using design thinking was established. The famous design firm IDEO emerged as a global leader of this movement. Design strategy consulting is not about just making an abstract theoretical strategy, as conventional strategic consulting firms do. Its strength is in the ability to utilize understanding of consumers to uncover new insights from their unique perspectives, and then synthesize these into a creative proposal for a concrete product or service. Currently, IDEO and McKinsey compete for strategic consulting projects.

What Are the 21st Century Skills?

To adapt traditional education systems for 21st century society, vast amounts of research has been done, with US leading the way. Since 2002, Microsoft, Cisco Systems, Apple, Oracle, Intel, and other corporations have cooperated with the US Department of Education in developing a list of skills needed for the future generation. In 2009, the UK, Finland, Singapore, and other countries with advanced education systems joined ranks, and the resulting 21st Century Skills are summarized below (ATC21S).

- Ways of thinking: Creativity and innovation, critical thinking and problem solving, decision-making, and learning
- Ways of working: Communication, collaboration
- Tools for working: ICT (Information and Communication Technology), and digital literacy

- Ways of living in the world: Citizenship, life and career, personal, and social responsibility



Among these, creativity and innovation together with critical thinking, problem solving, and decision-making, were selected as important thinking skills. Through this course of events, even top-tier MBA programs such as those at Harvard, MIT and Northwestern University have started collaborating with design schools to introduce programs that teach entrepreneurship and design. When Stanford University founded d.school for students of all departments to collaborate with innovation challenges, it was a global sensation. This trend of fusing business and design education is also occurring at the most well-established art schools and MBA programs in Europe, including in the UK, Germany, Northern Europe and Italy. While the West is already accustomed to hearing "I'm going to study design," in most Asian countries learning design is not common for business professionals.

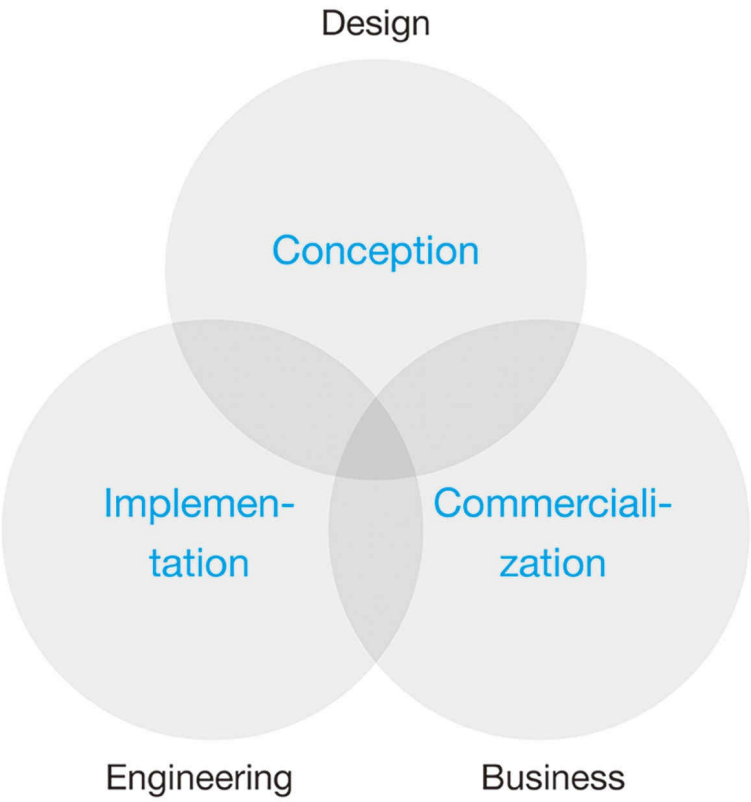
Merging the Three Hats of Design Innovation

To understand the design school curriculum, an essential starting point to support

innovation is the concept of the three hats of human centered design . This concept comes from IDEO's proposal that "New value is created when three elements — desirability, feasibility, and viability — overlap". I have rephrased this concept into the three hats of design innovation.

1.Conception: To conceptualize that which is desirable for people = The role of design.
2.Implementation: To realize and implement = The role of engineering.
3.Commercialization: To create a mechanism for society-wide commercialization = The role of business.

Three hats that support innovation



(Source: IDEO Human centered design, supplemented by author)

Design schools also ensure diversity of students through their admission strategy. Admitted students come from a range of backgrounds such as design, engineering, and business to consciously create curricula that merge all three disciplines. There is a philosophy that innovation is born from collaboration of small teams with a diversity of talents. At my school, designers and engineers, marketers and researchers, and other members of different backgrounds, learned and worked together on innovation projects. When designers learn business strategy, and businessmen and engineers learn design, they can become hybrid professionals who are capable of understanding and communicating with different styles of thinking. This leads to great diversity of thinking in collaboration and enables teams to ideate and implement innovative ideas quickly. I had little experience with the fields of design and engineering prior, but my marketing background helped me go into the design program as a professional of the business field, who would learn design for business and become a design hybrid talent.

Why Did a Marketer Choose To Go To a Design School?

I will first explain why a brand marketer like me, who mainly used logic in my job, decided to study design. After acquiring a bachelor degree in law, I started working at Procter & Gamble (P&G). I had to learn the P&G way of convincing people using logic, always explaining "the conclusion and the reason for it". From the start I was trained to thoroughly analyze consumer data, create a strategy, and convince management of its advantages and disadvantages. This is the typical MBA-style of managing business, and it enabled me to forge my marketing capabilities as a brand manager. At the time, for a young marketer like myself, data analysis and logical thinking were very powerful tools , for convincing people and getting promoted to a position of leadership.

Then A.G. Lafley, famous for the successful turnaround of businesses and cultivating innovation in organizations, assumed the role of CEO. A new skill requirement was added to logical thinking and strategic thinking in the list of "indispensable" skills for the organization: collaboration. When starting a new product development/marketing project, we would no longer work only within the marketing department, but were encouraged to create the project jointly with the R&D and design departments. The design department's participation brought particularly large changes. Product concepts were no longer expressed using only words; a world view engendering the product was visually expressed as a design theme, and shared within the team. The introduction of this process led to the sudden linking of concept, package and TV commercials. The stereotypical type of person that would succeed at P&G was the strong "president-like" visionary leadership type, who would push his own ideas ceaselessly; but the new changes also meant a change in the characteristics of leadership to foster more listening, collaborating, and motivating team members. Since the former kind of potent

leadership was not my strength, this change was like a windfall for me, and I also felt the range of possibilities made possible by collaborating in teams on innovative projects. In hindsight, this was my first encounter with design thinking.

How to analyze "intuition" using logic?

An important task of P&G's marketing is creating advertisements to attract more consumer purchases. This consists in selecting a scenario sketch, from a batch of proposals at an advertising agency, that both agrees with the product strategy and touches people's hearts. P&G established a training methodology for how to pick these advertisements, and one of the basic ideas behind it is "Believe in your intuition (gut)". Instead of starting out by considering reasons for or against, you train yourself to decide whether or not you like something on the spot. Then create the framework necessary to explain logically the reasons for your intuitive opinion afterwards. I was extremely poor at this. I could not decide whether I "liked" or "disliked" something using logic. When I started thinking "Proposal A has these merits, but proposal B has those merits," it would be impossible for me to claim "This is the one that I like!" I felt uncertain that I was doing things the right way, and when I thought about how my boss was assessing everything I said, I was uneasy mentioning my non-objective personal opinion. I was not able to sweep away this feeling of not being up to the task until I started having direct discussions with the creative director of our advertising agency. This was when I understood that my job was not to write pretty comments, but to discuss and clarify the context of what was important for my client and why. It was a good opportunity to learn that in the world of likes and dislikes, it is important to stubbornly spend time sharing your thoughts, even if they are not completely clear.

I also think this was when I discovered the idea of switching between a logical "left" brain mode, and the intuitive "right" brain mode. Before this experience, I did not even know where the button to switch between these two modes was

located. Note that "left brain" is a metaphor for logical thinking, and "right brain" is a metaphor for intuitive thinking."

Logic's Limitation, and Exploring the World of the Right Brain.

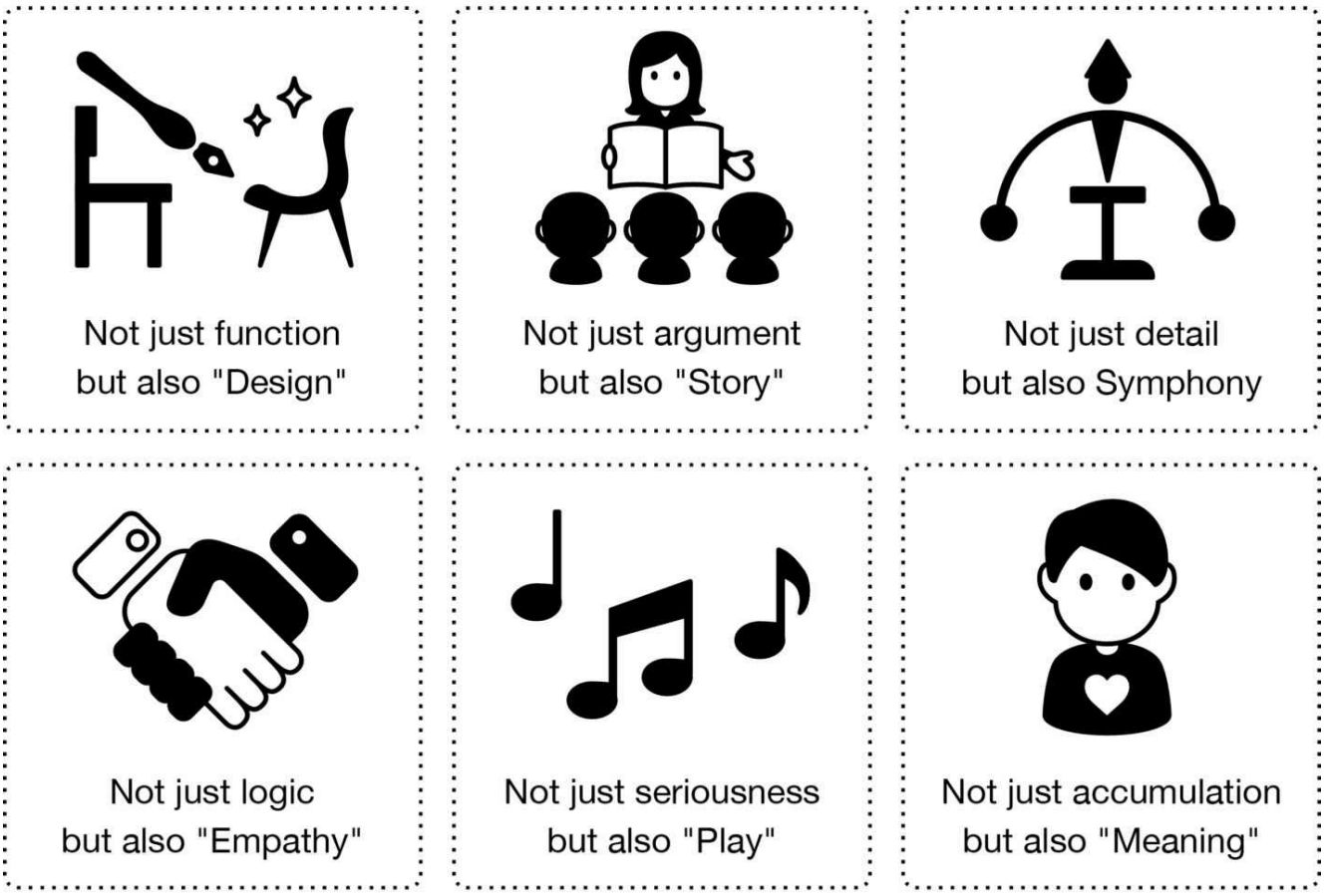
In P&G there are two types of marketers. The first type can achieve a V-shaped turnaround for a product whose sales or market share is declining, or may have the ability to launch a new product that changes the rules of the market. The second type steadily achieves gradual growth by using brand assets. I started feeling that there was a pervasive, essential difference between the first type of marketers, and the second type that I belonged to. Marketers belonging to the first type are not merely intelligent, but have the ability to imagine a future differently from others. On the other hand, although the marketer, of the elite team that launched Febreze, can make eye-opening proposals yet he never knows when he will suddenly get a good idea… At the time I did not know why these two types differ so much, but now I think it stems from how creative thinkers and logical thinkers use their brain differently.

I started thinking differently about my work style. Just keeping the conventional business running was not enough: I wanted to become the type of first-class marketer like my senior, who can give rise to innovation that changes the rules of the market. And so, I started exploring how to achieve this.

Encounter With the World of Creativity

While pondering these issues I came across the popular book "A Whole New Mind" (Daniel H. Pink, Riverhead Books, 2005). In this book, the 21st century is called the Age of knowledge and creativity workers. Six essential senses for people of this creative age are as listed: Empathy, design, story, play, symphony, and meaning.

6 essential senses for being creative



(Source："A Whole New Mind", modified by author)

When I read this book, I intuitively knew: "This is what I need to become an innovator!" To train these senses, I decided to try out a range of techniques including design, coaching and improvisational performance to explore this novel world. The book "Drawing on the Right Side of the Brain" (Betty Edwards, J. P. Tarcher, 1979) left an especially strong impression when I first encountered it. It claims that people who cannot draw are not lacking talent, but are not looking at things using the right brain mode. The only thing I needed to do was to change how I view the world. Since, according to the book, anyone can learn how to draw just by using the right brain mode, I participated in a workshop. By experiencing how to look at things with my right brain, I started pondering the possibility of not only training the use of my left brain, but also my right. I will provide evidence. The drawings below show the change in my self-portraits before and after the 5-day

workshop.

The difference in these two drawings resulted from just changing how I look at things. At the workshop we copied paintings by Picasso that were turned upside down, in order to practice looking at them as they are; not as famous paintings, but just meaningless collections of lines. This is what it means to switch to the right brain mode. Think of the trees that you see every day, lining the city where you live. By focusing on and becoming conscious of the tiny differences in their color and shade, you can switch from left mode thinking to right mode thinking.

The necessity of design thinking and collaboration in large corporations

Later I started working at Sony, where I helped start up a global marketing insight team, and took part in the company-wide product design process. There were more engineers itching to create new products at Sony than I had imagined, and I felt their energy was like bubbling magma in a volcano just about to erupt. There were many

people there with a creative mind-set, who wanted to put their work out for the world to see. The company culture "Don't rely on the consumer's opinion" also made a strong impression. Coming from a marketing culture, where we always were looking for ways to fill needs as they appeared to our customers, this was a big change. Locating latent needs that have not yet surfaced and giving shape to a concrete solution required a different kind of know-how.

The organizational structure of the Sony's Japanese businesses was also different from P&G. It was not unusual for the product development department to outnumber the department of my previous company by more than 10 times. With different departments having different points of views, I could sense the difficulty in aiming to create new value, and then unifying the direction the product takes. I started questioning what missing piece was needed for people with different viewpoints to create new value while always keeping the user's life context in mind. At this time, I discovered design thinking as a product development methodology for unifying the Sony's vision of customer understanding, research, product planning and design; and I started looking for a way to study this subject seriously.

Variety of Choices for Graduate School

However, it turned out to be incredibly difficult to find a graduate school for someone like me, who had never received any education in design. But when I one day tweeted "Are there really no graduate schools where a businessman can study design thinking?", I received a reply from a student at ID. My wish of going abroad to study turned into a realistic goal. Unlike MBAs, information concerning studies at overseas design schools is extremely limited. Just like me, the ID student had to put a lot of effort into researching information about possible target universities. Most art schools have a strong reputation as being the territory of designers and artists, and even if they have courses that merge business and design or art, the curriculum is different from school to school. After looking for schools

that met the two conditions “non-designers can apply” and “fuses business and design”, I narrowed my list down to two candidates: ID and Carnegie Mellon University. I started preparing for the admission. After visiting their campus, I found that ID is a graduate school oriented towards designers, but while a certain level of basic designer skill is needed, almost half of the students are engineers, marketers, researchers and other business professions. I chose to prioritize learning about design in a business setting, and decided to enroll at ID.

The courage to stand out

For a businessman to study at a design school is a nearly non-existent career choice in today’s Japan. For me, who lacked artistic talent and had been in the weaker half of all my art classes from elementary through high school, jumping head-first into the world of design required a lot of courage. When I told people I wanted to go to a design graduate school, most reacted with “Oh, that’s an interesting choice…”, and some even said things like “You’re already 30 years old, and you’re planning on becoming a designer? Really?” or “What about your career when you get back?”. Although this choice is slowly gaining acceptance, this was also the experience of all my friends who decided to attend design school, and we remain a tiny minority.

Throughout my studies at ID, I struggled with Japanese society’s particularly strong cultural antagonism towards those who stand out, and the tendency of people to judge me as if I was trying to show off by being “creative”. Exclaiming that I am a designer or that I am creative requires a lot of courage. Believing it is even harder, especially for me who had no experience or accomplishments to speak of. Troubled by such worries, my studies overseas started.

column

The History of the Institute of Design (ID)

Illinois Institute of Technology's Institute of Design (ID), where I studied, was founded in 1937 and is the oldest educational institution in America to teach design thinking. It was a pioneer in the field, it even established the first American doctorate in design. Today, 130 graduate and 10 doctoral students are enrolled there. In fact, the history of design education is intimately related to ID. As early as the Altamira wall paintings, design has developed as a means of communication between humans since primitive times. Even the paintings one can see in art museums, such as the Louvre, developed from a tradition that was originally a means of visual communication (think modern media such as newspapers and magazines). Even today, when hearing the word "design", many people imagine graphic design, which uses visual expression, or art, in attempts to question and express ideas. That is because these were the origins of design, and also used to be its most fundamental function.

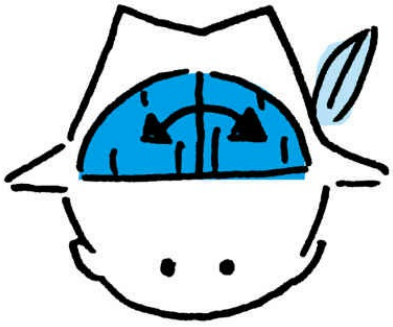
This trend changed with the arrival of the industrial revolution. The industrial revolution of the latter half of the 19th century led to the replacement of the craftsman, who would individually craft each single item, with standardized mass production. This resulted in products that might be functional and efficient, but lacking in beauty, flooding the markets. The antithesis to this development was formulated by the art school Bauhaus, founded in Germany after World War I. Bauhaus created a movement to propose the merging of rational and functional engineering with aesthetic art, in order to create something different from the functional but ugly products flooding the world after the industrial revolution. The underlying ideal was to move away from a mentality of "creating whatever products we can create", and to focus on design and beauty using human centered

design, which could then be mass produced. With this, the educational institution of Bauhaus aimed for the fusion of art and technology in industry.

Bauhaus had a large worldwide impact on contemporary design education, but with the rise of Nazi-Germany, Bauhaus institutions were closed after only a little over a decade. However, one of the founding members of Bauhaus, Moholy-Nagy, was invited to Chicago, and there he founded New Bauhaus as an educational institution for design in 1937. New Bauhaus became the foundation of present-day ID, and during its long history became the source of concepts such as human centered design, fusion of art and technology, and user centered design.Incidentally, the foundation of d.school was supposedly greatly influenced by ID's program.

chapter 1

Learning hybrid intellectual production from designers



"We are approaching the concept era, an era where society and the economy will be built by orginality, empathy, and the ability to see things all-embracingly. (...) Abilities that were previously regarded as trivial and insignificant"

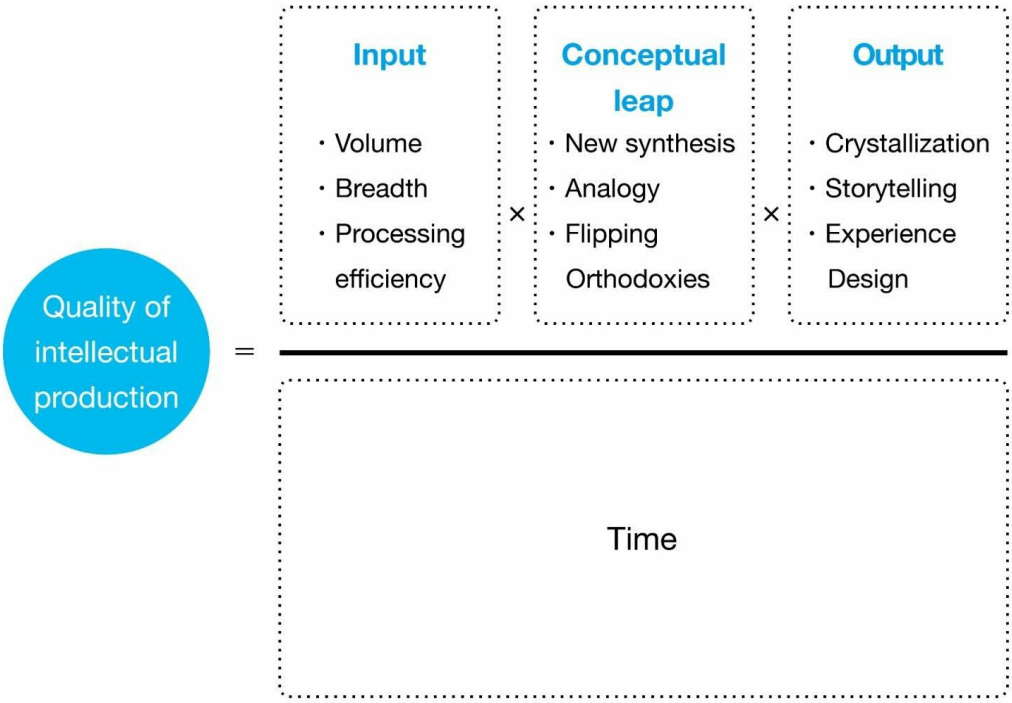
Daniel Pink (writer)

Designers' Know-How for Improving Creative Production

In order to master design thinking, the very first thing I would like to highlight is the thinking method that designers use. Some people say "I haven't come up with any interesting ideas, even though I followed the design thinking process to the T". But I think that many fail because they only trace the process, without changing their way of thinking. Linear thinking methods such as logical thinking and critical thinking are the most prevalent in the business world, but the thinking style that I learned at ID is a type of hybrid thinking that uses both linear and nonlinear thinking. This involves balancing the left brain's power of logic and the right brain's power of imagination to discover one's own unique perspective, and thereby turn creativity into one's own intellectual productivity.

In this chapter I have divided the essentials of this hybrid thinking process into three stages: quality of input, conceptual leap, and output. By making efficient use of your limited time to improve the quality of each of these parameters, it is possible to greatly improve your creative thinking capability. In addition, to help the reader practice the kind of thinking that produces new perspectives, I supply a right brain thinking method unique to designers that I learned by collaborating with my designer classmates. These are some common questions that designers tend to ask themselves and ponder on during their work. The questions are found at the end of each section, so please try and think through them for yourself.

Breakdown of the designer's thinking method



Input: Collecting Visual Stimuli and Thinking Visually

When designers start a project, they conduct inspirational research in order to come up with new ideas. When conducting research in the world of business, the standard practice is to surf the web or search articles to collect large amounts of data or facts and save them in an Excel spreadsheet. This data is then used to determine if the market is expanding or shrinking, to figure out a competitor's strategy, or to find changes in consumer/market needs. Designers also conduct this kind of research, but an even more important practice is finding input to stimulate their inspiration by using visual images. This can entail gathering visual data such as pictures and video, or personally going to see a location for yourself.

Let us say your mission is to design a completely new clock. A designer might search Google Images for clocks with novel shapes, search magazines for lifestyle scenes including clocks, or pick out YouTube videos with clocks. Following this they might visit places where clocks are sold, or interview clock aficionados. Other sources of inspiration may include browsing Kickstarter products or art exhibitions. None of this information is considered to be "objective information" in the world of business. Visual information, however, actually contains overwhelmingly more information than textual information. Finding new hypothetical perspectives by imagining human emotions and lifestyle scenes as concretely as possible is the defining feature of a designer's research.

Get Out of the Office

Rather than processing condensed secondary information such as text, it is important to gather lots of visual information, physically visit and observe locations and conduct interviews in order to immerse yourself in and absorb primary information. When you conduct research with a designer, you notice how

much information they use as input. Taking the example with the clock, it would be commonplace to gather fifty or a hundred photographs of clocks from image searches. By going through this process, and by taking in a large amount of concrete experiences and images, it is actually possible to gather hints as to what kind of output you are trying to achieve. Design research may include visiting the residence of a user, and it is common to take 200 or 300 photographs during a single visit.

I've also witnessed brainstorming sessions where each participant would prepare fifty to a hundred pictures from sites such as Google Images or Pinterest. Images, videos, and personally experiencing a location; all of this involves huge amounts of information, and it should be possible to make new discoveries that ignore fixed notions or prejudices.

A designer's thoughts concerning this section can be explained using the following questions.

Thinking

What kind of visuals might be helpful in developing a concrete design? How can I gather as much information as possible?

Experience unfamiliarity

By experiencing unfamiliar environments, you can expand your range of perspectives. Below are examples of axes that can expand your perspective of design research in the discovery stage of an innovation project.

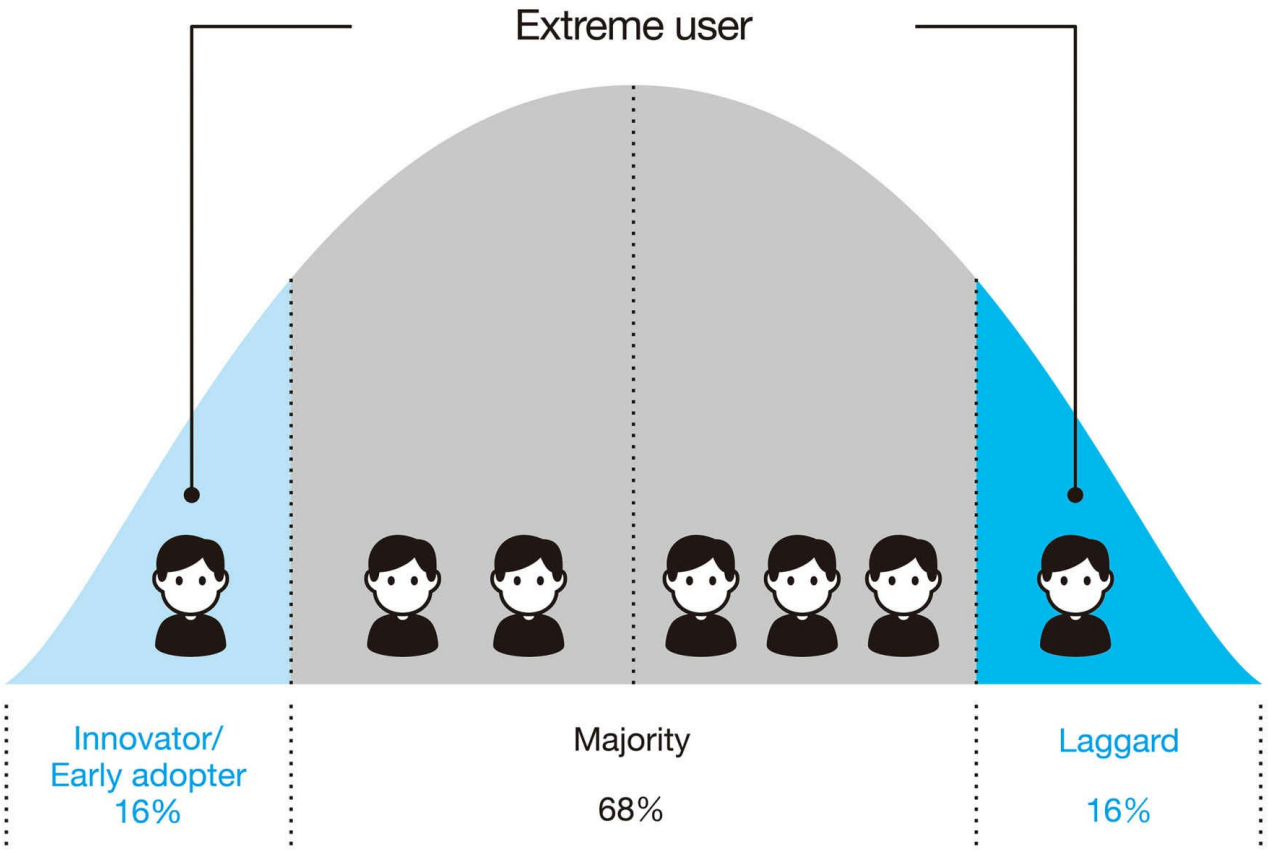
- Human axis: Experience the lives and livelihood of people in environments completely different from your own

- Area axis: Exposing yourself to examples from areas of work/study with some common features, but completely different from your own.
- Geographical axis: Exposing yourself to things happening in completely different parts of the world.
- Time axis: Learning what changes, and what stays the same throughout the ages from a historical perspective.

A common example of traversing the human axis is to intentionally pick "extreme users" (users with extreme preferences) as observation targets for user research in an innovation project. In the world of business, when conducting market research, it is common to pick average users as market representatives. However by intentionally picking extreme users, for example, heavy music users such as DJs, or young people who buy cassette tapes, it is possible to expand the breadth of your thinking.

The geographical axis is also important. When the global design firm IDEO conducts design research, they research a single theme in several different areas of the advanced world. When they have a project to design a new type of home, they may choose to conduct field work on new housing complex concepts in a country at the forefront of community construction, Denmark, or on concentrated housing in the tiny territory of Hong Kong.

User distribution according to diffusion of innovation theory



Using historical analysis to conduct design research is also very powerful. When working on a research project for electronic books, for example, you might study the history of reading in order to understand the experience itself. By going back to Gutenberg's printing press of the 16th century, or as far back as the age of hieroglyphs, and looking at what is common throughout all of human history and what has changed through the ages, you can search for that which is missing in our current age and that which will be more needed in the future.

Thinking

What kind of place can you visit to find viewpoints different from your own?

Thinking without words: Visual thinking

Once you start practicing design thinking you will have to process large amounts of information, but there are tips to help you digest this information more easily. Making a habit of doodling on post-it notes and other note-taking strategies are certainly important, but techniques that involve summarizing information visually are also incredibly powerful. One of these techniques is so-called "visual thinking", which involves thinking and understanding visually. It is actually possible to practice visual thinking just by changing the way you take notes. Below figure shows a page of notes that I took during a lecture on user observation in the second half of my studies at ID. The illustrations are not copies of the teacher's blackboard drawings, but my own imagination of what the teacher explained verbally, summarized as illustrations.

The Back of the Napkin (Dan Roam, Portfolio Hardcover, 2008) is a very popular visual thinking textbook intended for designers without an arts education (http://www.danroam.com/the-back-of-the-napkin/). The title refers to the practice of taking notes on the back of the napkin in a restaurant, and suggests that when you're already taking notes, you might as well make the contents visual, rather than

only written text. The Back of the Napkin was written to explain to businessmen the visual thinking techniques that originated at Stanford University. This book has become a must-read in the American design industry.

By making a habit of drawing illustrations, you will find yourself first attempting to understand the information, and then translating the logical structure of it into visual information. By repeating this process every day you will slowly be able to use both the left and right sides of your brain for their proper purposes. By converting words into a concrete image, you will also gain deeper understanding.

In addition, when you make a habit of expressing data and ideas as illustrations, you will start thinking about the relative positions of the illustrations as well as the overall layout, and in complicated problems it will be easier to understand what items should take priority. You can practice this by creating illustrations of what people are telling you. In my case, I started by making schematics during lectures, and when I started feeling comfortable with my drawings, I also began drawing illustrations during meetings, even while I was thinking or talking about other topics. Among designers and strategy planners I often see people who almost never use text, but take all their notes with illustrations and charts.

Thinking

How would you express a discussion with an illustration?

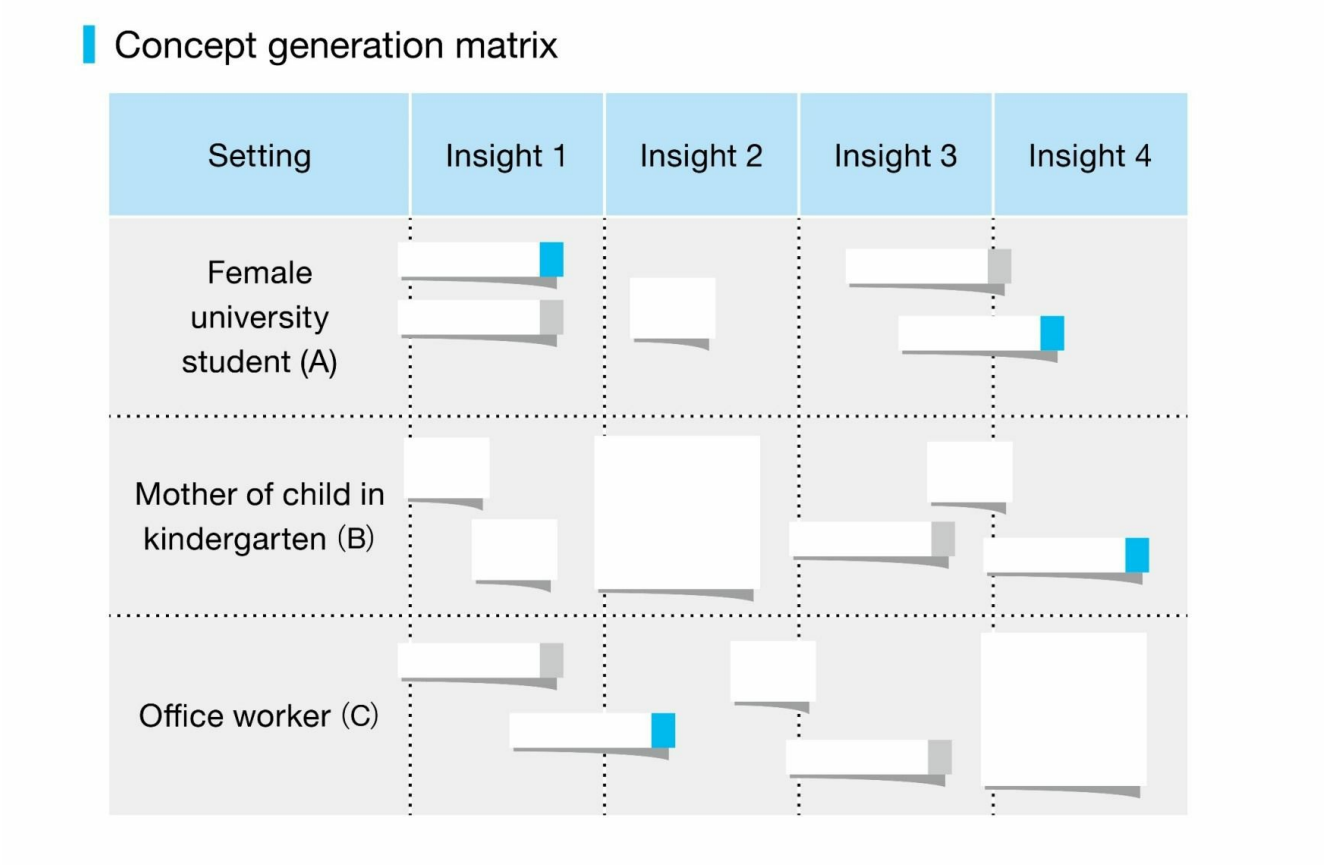
Conceptual Leap

After taking in a range of inputs, from the obviously important to the seemingly irrelevant, we must advance to the next process: making our ideas take a conceptual leap to discover new and unique perspectives. There already exist many ideation methodologies, but finding completely new perspectives is incredibly difficult. The process is also difficult to understand, since you never know when the leap will take place. This way of thinking is one that a business professional almost never applies in his regular work. Even during a brainstorming session to generate ideas for a project pressed for time, many people experience a lack of results. One thing I learned at ID through classes and observation of designers is that there is an established process for even this seemingly incomprehensible phenomenon, and that it can be mastered with practice.

This process can be roughly divided into the following three design thinking methods: new synthesis by connecting different kinds of combinations; analogical thinking by finding similarities between seemingly different things; and flipping orthodoxies by changing rules that are taken for granted.

Finding Ideas with the Forced Ideation Matrix

The new synthesis process involves discovering new ideas by forcibly connecting elements that are seemingly different. This ideation methodology, which makes use of many different connections, is taught in a subject called “Synthesis” at ID.



One conventional method is to make a concept generation matrix. The vertical axis lists hypothetical user segments, the horizontal axis lists user needs, and new ideas are generated by making new combinations. Connect large user segments that have been determined through research, with the needs of the users of each of these segments. Then you can visually draw the solutions that users desire in a brainstorming session.

A businessman may question the necessity of contemplating seemingly meaningless connections, but given a lot of trials, many new perspectives will spring from completely unexpected combinations. When you get used to it, you'll start seeing possible novel combinations everywhere even without writing down these axes, like "education x design", or "fishing x wearable devices".

Thinking

Can you think of anything new by using new combinations?

Analogy: Finding similarities between completely different things

One of the designer's most powerful ideation tools involves detaching from their current fixed point of view (mental model) and thus intentionally making a conceptual leap. This method is called analogy, and it also makes use of metaphors. Analogy refers to pointing out similarities between things that appear different at first glance. In "An Inconvenient Truth", Nobel Peace Prize winner and former Vice President of the United States, Al Gore, used the phrase "The planet has a fever" to raise the problem of global warming. This is a good example of analogy use, as it compares global warming to having a high fever. You can conceive new ideas by connecting knowledge of the familiar world with knowledge of a world you know less about. These new ideas can be conveyed to someone else by connecting them with things that the listener already knows well. It is, in other words, necessary to go back and forth twice between the familiar world and the unfamiliar world. This conceptual coming and going leads you to look for similarities between things that look completely different, and this is what analogy thinking is all about. In order to practice analogical thinking, ask yourself "What does this resemble?" and "What is the metaphor of my new idea?". A familiar way of structuring metaphors is "A is like B". For example, "Design schools are like ateliers where you draw the future." When you want to develop this into analogical thinking, you just need to think of the similarities between A and B.

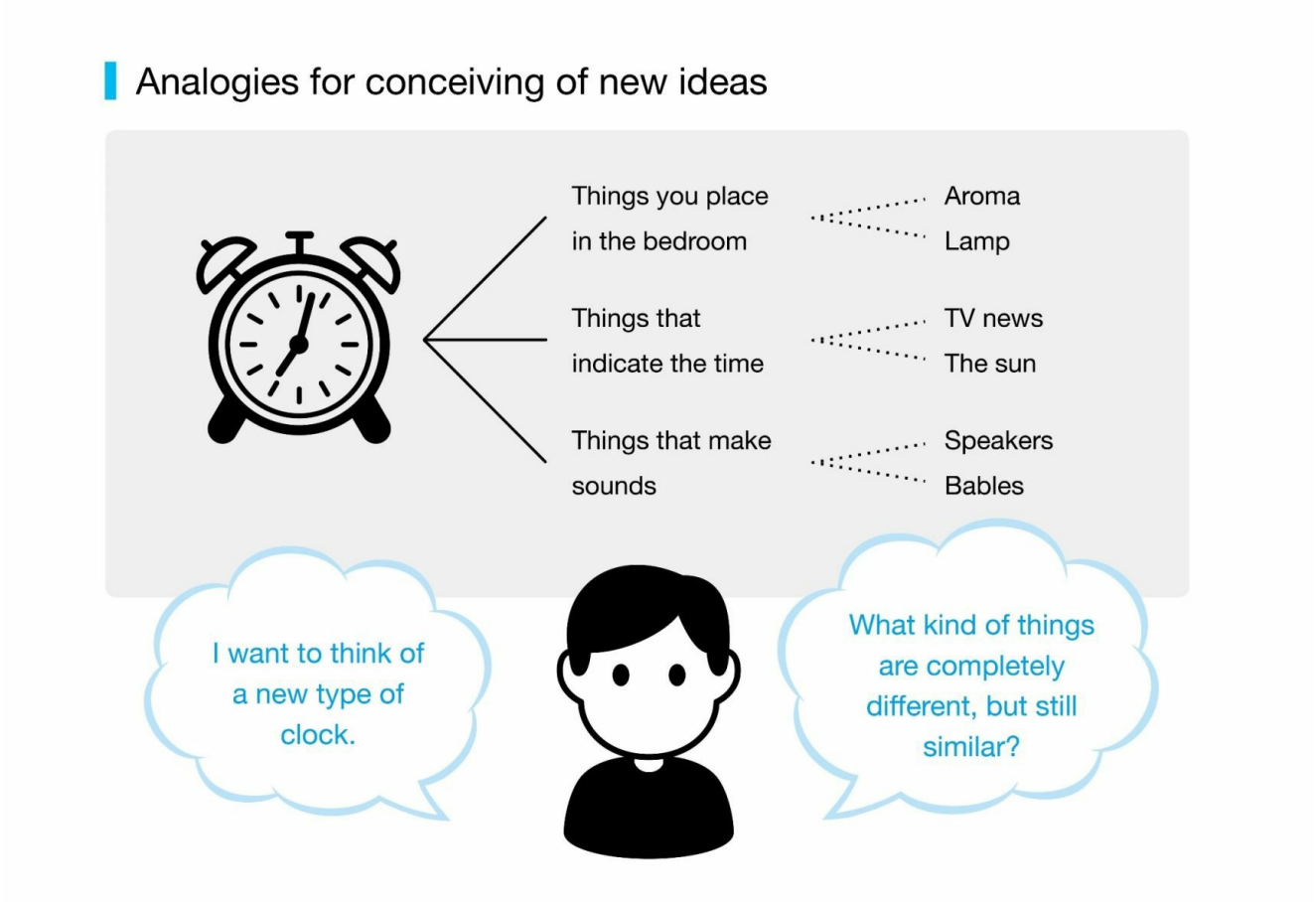
Analogy is a skill that was particularly difficult for me to find my way around, but there are many experts in the uses of metaphor among designers. At design school I discovered a very important point for acquiring this skill. When trying to make a conceptual leap, rather than thinking using words, it is more effective to search for similarities in pictures from magazines, or other methods that force you to think visually. Words are abstract concepts. Thinking using only words makes it difficult to find similarities that lie in details, and it takes much practice to think metaphorically this way.

The analogical thinking that designers are practicing unconsciously every day can be roughly divided into the following two categories:

1. Analogies for conceiving new ideas
   (Make strange familiar)
2. Analogies for conveying ideas
   (Make familiar strange)

Analogies for Conceiving New Ideas

By taking the theme or problem you are grappling with, and thinking about what the result might be if you likened it with something completely different and put this in its place, you will discover unintended connections and essential challenges that they share. And by forcibly connecting it with something different, you will expand your imagination. One way of practicing this method that I can recommend, is to select a theme and then go for a walk in the city and take pictures. For example, if your theme is to find a new type of eraser, you would be walking around the city looking for things that remind you of erasing, and taking pictures of them. You could also try to look at image sites like Pinterest and Instagram, and collect pictures that somewhat resemble what you are looking for.



Below figure is a photograph of a garden in a Kyoto temple. If you look carefully at this photograph, you will find that it includes a great variety of elements.

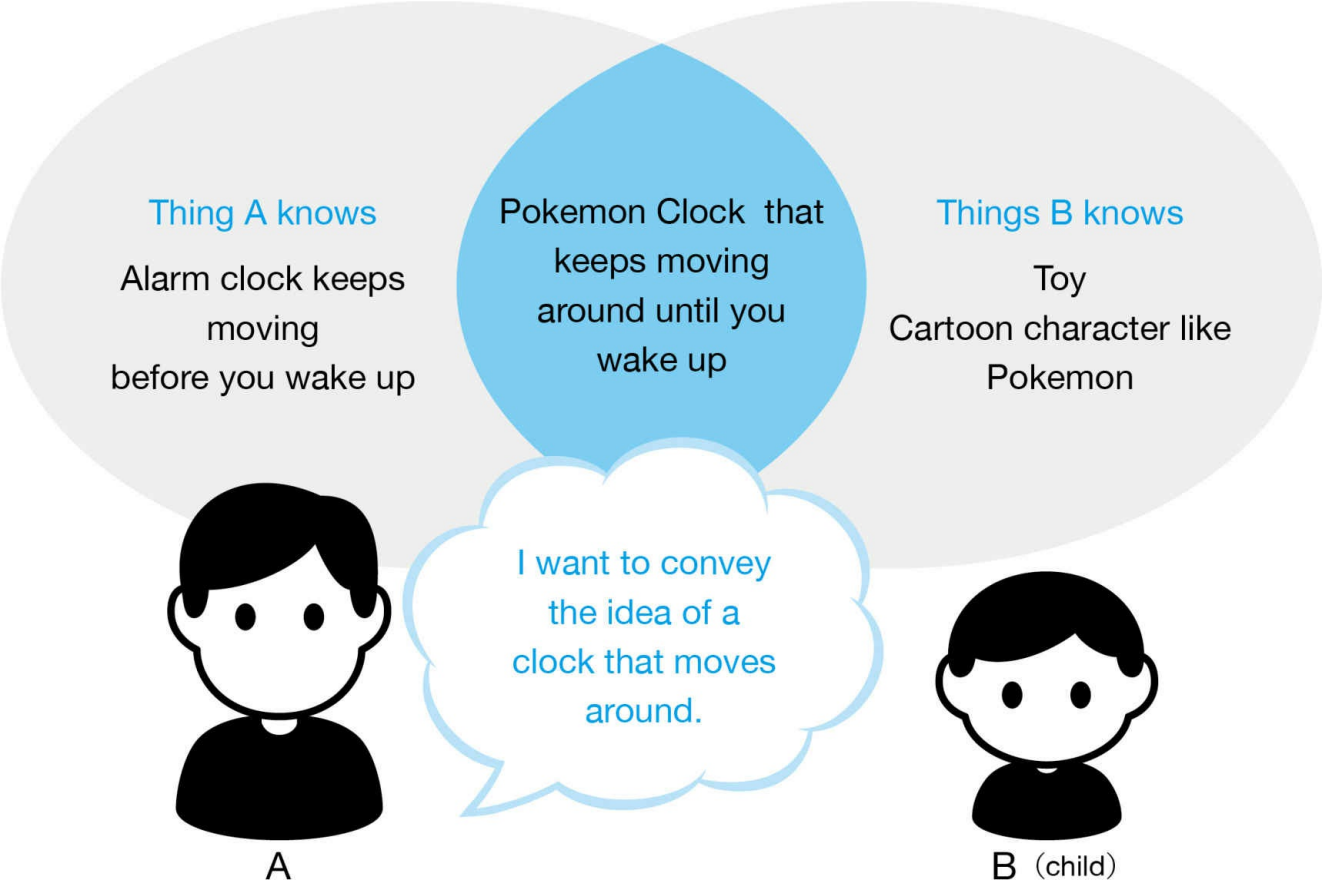
- A place surrounded by woods
- A tourist spot
- Tranquility
- A lake that is the dwelling of various lifeforms
- The water surface is like a mirror
- The water surface is as opaque as cocoa …

After extracting these elements, try and think if you can liken any of them with one of the services you are thinking of at the moment. What kind of clock would let you feel tranquility? What kind of clock is like a mirror? What kind of clock attracts a lot of people? The possibilities are endless. There are tools in the form of cards that help you generate random new thoughts using abstract visual metaphors. Tarot cards, with their abstract illustrations, are perhaps one example of this.

Analogies for Conveying Ideas

This section focuses on how to use analogical thinking as a communication technique for conveying new product and service concepts, by comparing them with familiar concepts. Since people cannot understand things they have never heard about before, it helps to have a comparison to what they already know. The biggest difficulty with innovation is how to convey novel ideas in the most appealing way possible. Novel ideas are not familiar to most people. If no one has ever seen something before, it is no surprise that it will be difficult to understand. This is when you use metaphors. When iPod went on sale, the slogan "1000 songs in your pocket" was used. This explains perfectly the novel experience of fitting your entire CD collection inside your pocket. One way to practice using analogies for communication, is to try and explain your new concept to someone who has no knowledge of that field, like a child, or your mother. When you try to explain, you are naturally forced to use analogies that he or she already knows, and this lets you polish your ability to use analogies.

Analogies for conveying ideas



Presentation methods that involve drastically decreasing the amount of words and using abstract illustrations in their place, have become very popular recently. This way, using visual analogy thinking can also be very powerful when doing presentations. I will not go into more detail here, but I believe you can develop your analogy thinking by keeping in mind the following questions during your daily work.

Thinking

What things appear completely different, but are similar in some ways?

What would you compare this to if you had to explain it to a six year old?

Creating new ideas by flipping orthodoxies

It is difficult to suddenly, all by yourself, come up with great ideas that make existing business obsolete, so-called disruptive innovation. This is because you are unconsciously taking common practices and ideas for granted. Are there any methods for coming up with ideas that break down what seems obvious?

Imagine that we start a public service design project of coming up with a new model for mandatory education services. First, we will write down a few implicit, generally believed rules/myths concerning current education. For example: "Education is more effective in small classes." "We are moving from physical to online classrooms." "The impact of education is proportional to the ability of teachers." "The core of education is excellent teachers."

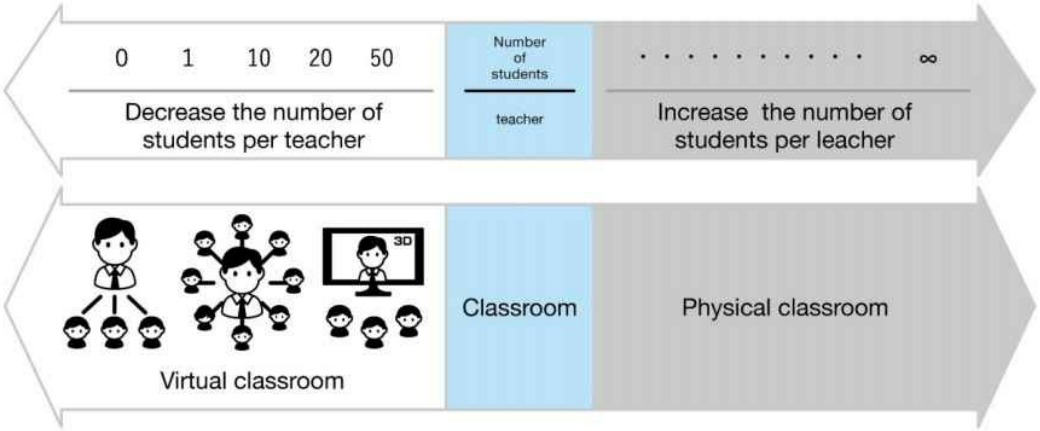
Next, we will write down the opposite of each of these myths. "Education is effective in large classes." "Physical classrooms will become even more important educational services." "Educational services without teachers." The opposite of myths are obviously things that are not believed right now. These are perspectives from which innovation may be formed.

Example: Create a new concept of education

1 Clarifying existing Orthdoxies (implicit assumptions).

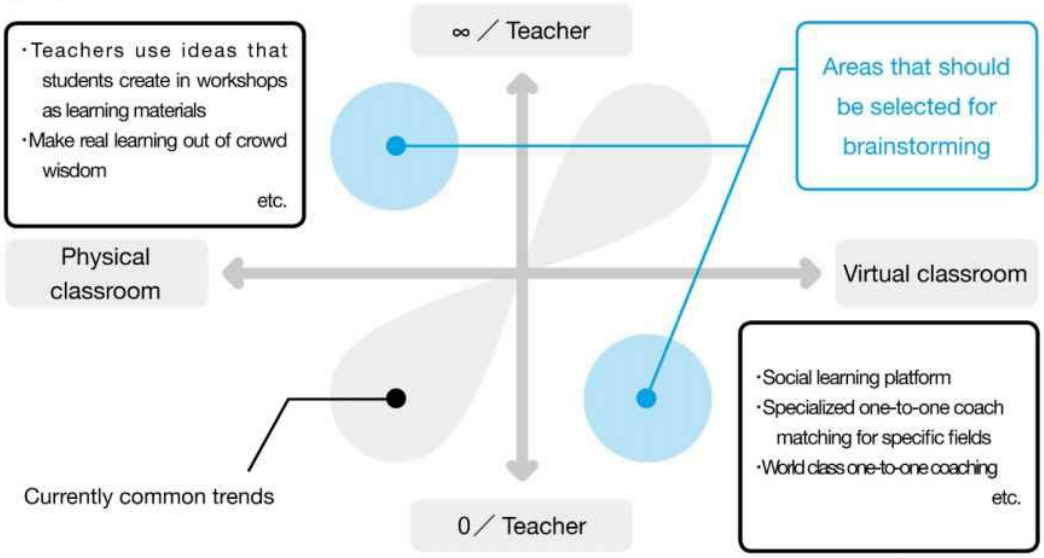
- ☐ It is important to decrease class size in order to improve learning efficiency.
- ☐ We are moving from physical to virtual classrooms.
- ☐ Children play around too much so we should increase the time they spend in school
- ☐ Teachers must have excellent teaching skills.

2 Create two axes and think in the opposite direction.



Change the combination of axes and repeat

3 Narrow down to unexploited areas



Out of this we take two axes, insert adjectives at each end, and make a 2 by 2 diagram such as the one in above figure. In this diagram, we can say that “efficient education is possible even with many students per teacher” x “education in a physical classroom” or “virtual classroom without teachers” are novel, and this might give birth to new perspectives. While brainstorming, we might as well narrow it down to relatively unexploited areas, and thereby come up with ideas that suggest new perspectives. Concrete ideas that may emerge from this are, for example, classes taught in a physical classroom but using teaching materials created in real-time on the Internet with the wisdom of crowds, or online social learning platforms to make it easier to learn together with peers. Using this method, you may also conduct brainstorming sessions where you list existing business rules (practices and ideas taken for granted), and then write down their contrarian positions.

Thinking

What “common sense' ideas” are taken for granted? What novel axis can we use to break down these preconceptions?

Output: Using Experience Design to Appeal Directly to Emotions

Expressing what you want to convey by condensing and simplifying your ideas as much as possible, and creating a dramatic experience in a form suitable for the recipient, will result in an emotionally moving presentation. These are functions that are performed subconsciously even in normal business situations.

Crystallization

In order to crystallize various ideas into something simple, designers use different formats such as single page posters, naming, slogans and so on. By intentionally imposing format constraints, for instance by limiting your expression to one page, you create an environment where unnecessary items are automatically trimmed off.

During our user observation course we spent two months researching and analyzing, but one day our teacher suddenly said, “Alright everyone, now you will draw on the whiteboard a prototype of a poster summarizing your research results. You have five minutes.” It may sound incredibly difficult to condense the results of research conducted over a long period into only five minutes, and in poster format at that. However, since we were faced with the constraint of summarizing everything on the whiteboard in a very short amount of time, I think it turned into a good opportunity to discuss and bring to light the very essence of the results. By making the prototype poster, the essential structure became clear at a glance. Posters also allow you to express your visions by changing background colors and templates, and they are an excellent format for condensing information.

MIT Media Labs’ Hiroshi Ishii often says during his workshop things like “Explain your idea in 140 words or less, like when tweeting”, or “Express yourself in four Chinese characters” for Japanese and Chinese.. By using a format that

condenses your ideas, you can shave everything off until only the essence remains. By using right brain thinking to consider visual expression, layout, and copywriting into a format using only one page, it becomes possible to simplify using the designer's tool of subtraction. While using your right brain, try to notice the difference between this and everyday left brain thinking, characterized by picking out the necessary parameters and then arranging them by priority.

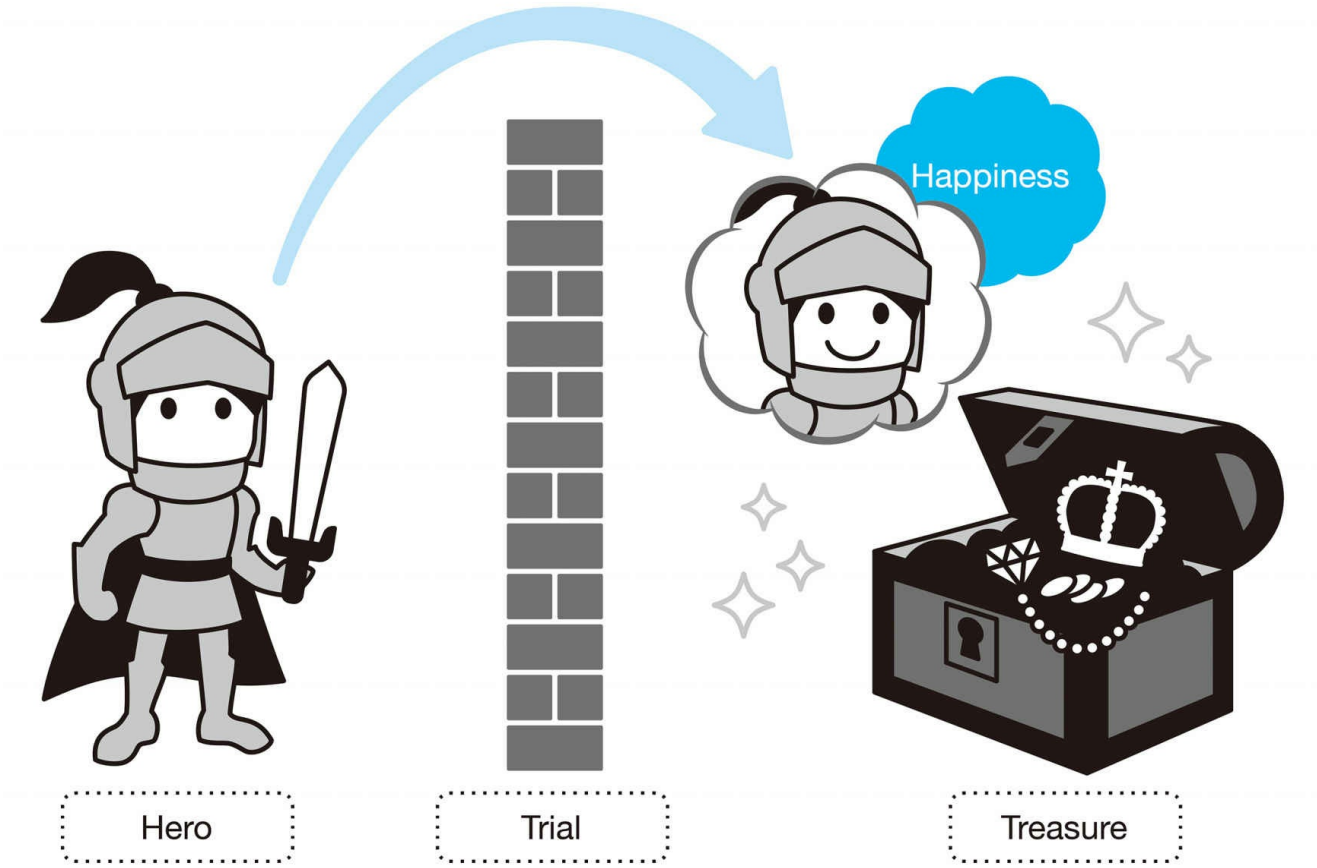
Thinking

What simple format would enable you to condense your ideas and express them?

Storytelling

Designers' presentations are characterized by trying to express a story as concretely and emotionally as possible. For example, when presenting new user insight from design research they will not introduce an objective list of facts of what is common between eight different people, but rather tell a story about one characteristic user. This enables the main point of their presentations to move people's hearts. Here, the framework known as "the hero's journey", often used for making Hollywood blockbusters scripts.

Basic structure of the hero's journey



Very simply put, we replace “the four elements making up the basic structure of a story” with “the four elements making up the basic structure of the service we propose”, with the user as the hero. The four elements making up the basic structure of a story:

1) The hero
2) acquires a treasure
3) by overcoming an obstacle,
4) and thereby becomes happy.

The four elements making up the basic structure of the service we propose:

1) The hero, who is the service’s user,
2) acquires the benefits of using the service
3) by grappling with everyday challenges,

4) and thereby becomes a little more happy.

The first step in making this is to write a log line; 2-3 lines expressing the gist of the story. They are similar to the short summaries written in film pamphlets and on the back of book covers. The log line of the film Schindler's List would be something like: "The main character, a playboy in the manufacturing industry, saves 1100 Jews from certain death. Shocked by the brutal actions of Nazi Germany, he fools important Nazis and turns his factory into a safe haven for Jews. Based on the true story of Oskar Schindler." On a side note, the countless film projects brought to Hollywood are supposedly first screened using only this log line. This illustrates how important log lines are.

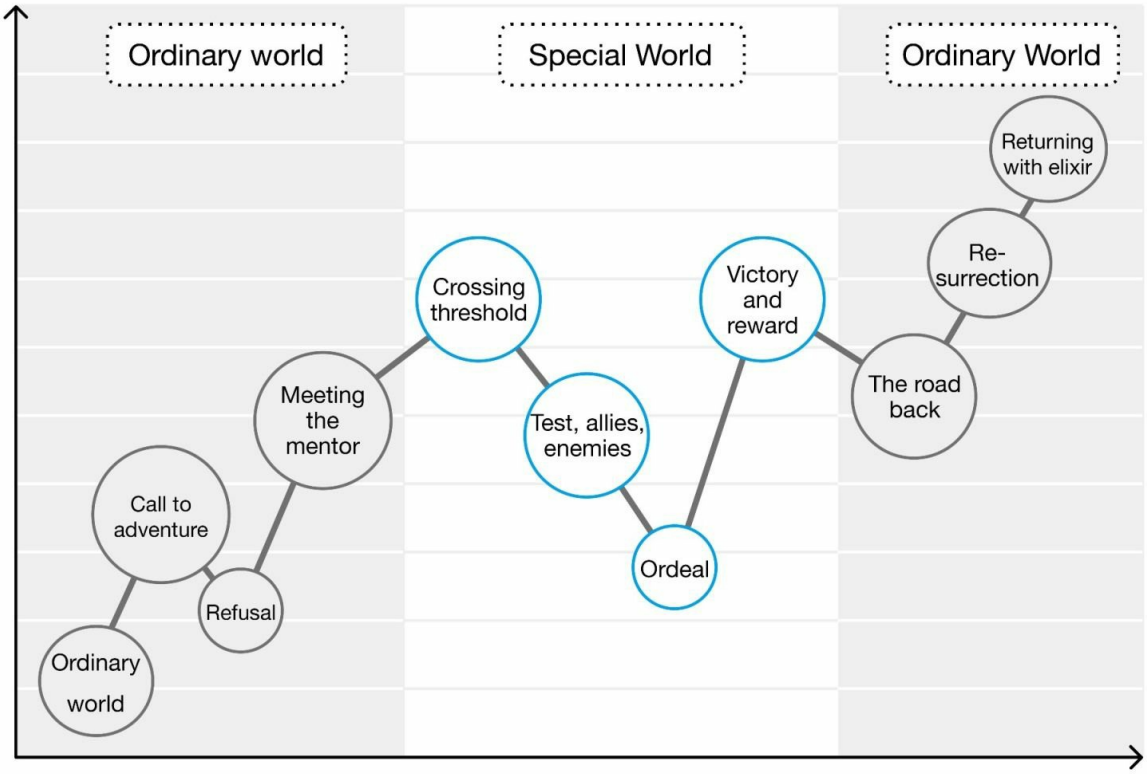
Once you have the gist of the story, its basic structure can be developed. To create the story, one can use the "hero's journey" as a model.

1) The ordinary world: What challenges are there in the real world?
2) Call to adventure: The occasion for leaving the every-day comfortable and uneventful world.
3) Refusal: The conflict between what can be gained (benefits) and what can be lost (costs) by embarking on an adventure.
4) Meeting a mentor: An unexpected supporter appears.
5) Crossing the threshold: With the help of a mentor, the hero decides to leave the familiar and ordinary world.
6) Enemies/ordeal: Hurdles encountered during the adventure that need to be overcome (competition, psychological stoppers, etc.). The hero is overwhelmed, but has the courage to overcome them.
7) Victory and reward: The reward (benefits) earned through overcoming the trial.
8) The road back and reaction: Lessons learned during the adventure, noticed

upon returning to the real world (human growth etc.).

9) Returning with the treasure: The state of success.

The hero's journey story framework



By writing one or two lines about what happens to the hero in each of the steps in the framework in the above figure, you can summarize the gist of the story. The key here is to be as specific as possible, and make good use proper nouns, adjectives, and adverbs that evoke images to make a lively story in a short text. By using this framework, it becomes possible to vividly narrate the perception given off by a product or service using a story.

Thinking

What stories or anecdotes can I use to convey my idea with the most impact?

Experience design

When designers present their ideas, they are not just good at making presentations, but also good at thinking of how to create experiences using many different types of

media. Ideas can be expressed not just with PowerPoint presentations, but also with posters, videos, or prototypes. Methods of expression that are redesigned to make a larger emotional impact include improvisational role playing, games, and life-sized mock-ups. The method should be selected according to the contents, to create the most suitable presentation.

When the design firm IDEO does consulting work, a communication designer is part of the team. The communication designer's only job is to think about what format should be used to convey the message in the final project presentation. One communication designer told me that "PowerPoint presentations are too obvious and not interesting enough. When considering what method of expression to use in order to convince my listeners, I think from scratch".

Thinking

Apart from PowerPoint, what is the best format or method to convey your ideas?

Underlying Premises of Design Thinking

So, how can you start using these thinking methods? People who predominantly use their left brain (common among businessmen), need to set aside several familiar habits in order to practice right brain thinking.

Try Not to Process Every Single Piece of Information

I already explained how sudden conceptual leaps can follow from going beyond familiar thought patterns while flooding your right brain with huge amounts of information. A common obstacle at this point is being unable to properly process all of the information. When handling large amounts of visual information, which is hard to analyze completely, it is impossible to process it in a way that would please everyone. Therefore, it is necessary to get rid of overly diligent attitudes that require you to summarize all information completely.

Try Not to Fear Uncertainty

When practicing right brain thinking, thoughts will often make leaps that can hardly be explained using logic. As your thoughts diffuse in various direction, the overall picture becomes harder to grasp. We cannot and even should not clarify all the thinking and steps and expected output. You may even think you are not outputting anything. But the more you think that you need to explain everything clearly, the less you will experience leaps of thinking. It is necessary to accept states that may seem chaotic and uncertain.

People Who Create Change are Using Design Thinking Unconsciously

The way I see it, one important reason for left brainers to learn design thinking is that they may become capable of using left brain-right brain hybrid thinking, a powerful skill for finding perspectives other people cannot see. Necessary skills for hybrid intellectual production, such as empathy (user observation), symphony

(synthesis), design (design in general), and story (storytelling), largely overlap with the six senses from A Whole New Mind that I introduced in chapter 0. Those with the know-how to produce intellectual output by using both their left and right brains can become very wise people with the ability to continuously create new innovation.

I believe one should divide one's work into two parts: creating new value, and business as usual. For the work that creates new value, it is necessary to dedicate time for using design thinking to generate new ideas. As for me, I use design thinking whenever I have a project I am excited about, or a project with great potential that involves making entirely new perspectives into something concrete.

Designer's point of view

- ☐ Research: search for inspirational visuals
- ☐ Thinking: use analogies to make a conceptual leap
- ☐ Presentation: gain sympathy by telling a story which leaves an impression
- ☐ Summary: express ideas visually by using only one page

Businessman's point of view

- ☐ Research: gather facts
- ☐ Thinking: think logically using a logic tree
- ☐ Presentation: use logic to persuade
- ☐ Summary: make an executives summary with three main points

column:

Overview of the ID Program

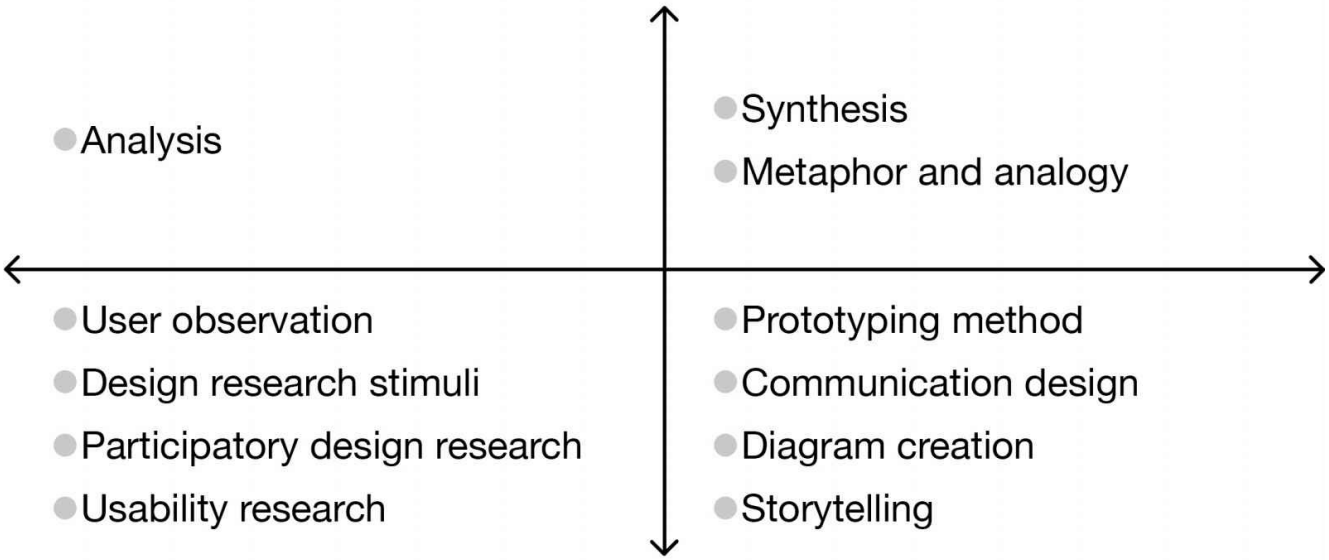
ID has two graduate courses: a 2-year Master of Design course, and a 1-year Master of Design Methods course (As of 2013). The Master of Design course aims at creating hybrid professionals excelling in both design and business. Additionally, students with no previous education in design take an additional year of basic photography, basic product design, basic interaction design and other introductory design courses, making it a 3-year course. The Master of Design Methods course that I took is similar to an "Executive MBA" course intended for students with more than 10 years of work experience. The contents of the lectures are the same as for the 2-year course, so it is a constant struggle to keep up over this short time period.

During my studies I also made friends with MBA students and students in other fields. Through our interactions, I felt that one important characteristic of design schools is the large number of team projects with the goal of outputting design. Of course, participation in discussions during lectures is also evaluated, but the quality of the actual output of the projects conducted with teammates every week are very important for the final evaluation. In addition, since the importance of researching user needs is stressed as a prerequisite for doing design, field work such as interviewing and observation are very frequent. In one year, I interviewed more than 100 American interview subjects. In hindsight, the experience of getting through these 100 interviews was also very important in allowing me to gain a much better understanding of the diverse lives of Americans residing in Chicago, and to really get a strong sense of the various values that people hold. The courses consist of programs with lengths varying from seven to fourteen weeks, and each program follows the same flow: research, research summary, design development,

and presentation.

Example courses at ID

Module for learning the design thinking process



Practical courses by subject

- Service design
- Innovation space
- Innovation strategy
- Design consulting

At ID, a framework for what can be utilized in practical work for design thinking projects is properly defined as in the illustration, and from this a variety of courses have been constructed. Of the six senses that Daniel Pink presented in A Whole New Mind, the four senses of empathy (user observation), symphony (synthesis), design (design in general), and story (storytelling) can be studied at ID. In addition to the above subjects that aims at systematically teaching the necessary skills to execute a design project, there were also subjects that involved workshop-style practical design projects with actual clients such as companies, NPOs, and governmental institutions. Examples of workshop subjects in 2012 included the following:

- Creative co-working space research project (for a large manufacturer of office furniture)

- Service design project for improving municipal office services (for Chicago city)
- Communication design project (for an Indian NPO)
- Service proposal project (for a venture incubation group in Chicago)

There are also possibilities to participate in research trips to countries such as India and Brazil during the winter or summer breaks, and thereby experiencing developing countries that are gaining increased attention in the design and business world.